

PLL-Based Wideband Frequency Modulator: Two-Point Injection Versus Pre-Emphasis Technique

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Abstract—This paper analyzes and compares two popular methods to widen the modulation bandwidth of a phase-locked loop, i.e., the pre-emphasis and the two-point injection technique. The analysis reveals that both architectures have the same sensitivity to gain errors and nonlinearity in the loop, though, compared with the pre-emphasis, the two-point injection scheme features less sources of error and does not require a phase detector with wide range and tight linearity requirements. The verification of the analysis as well as the comparison of the two modulation techniques is carried out on an accurate time-domain model of a 60-GHz digital phase-locked loop, taken as a case study and used to generate wideband chirp signals for a radar system.

Index Terms—Digital PLL, mm-wave, FMCW, radar, pre-emphasis, two-point injection.

I. INTRODUCTION

FREQUENCY-MODULATED continuous wave (FMCW) radars are widely employed in the state-of-the-art automotive sensors for identification of the distance, speed and angular position of the objects. However, the high cost of these systems limit their application to the high-end market. To lower costs, but at the same time, add a higher degree of reconfigurability and achieve a finer spatial resolution [1], a large effort has been recently devoted to develop fully-integrated solutions for radar sensors.

One key component of FMCW radars is the chirp generator, that is a local oscillator whose frequency varies as a sawtooth or a triangular waveform spanning several GHz with tight accuracy [1]. Conventionally, chirp generators are based on phase-locked loops (PLLs) with incorporated frequency modulation capability. Unfortunately, the limited bandwidth achievable with a PLL introduces a low-pass filtering in the modulation path. This produces linear distortion and modulation errors which degrade radar performance.

Several recent works on PLL-based FMCW modulators employ the so-called *two-point injection* technique to over-

come the limited bandwidth of a standard PLL [2]–[4]. The technique is based on the additional injection of the modulating signal in a point of the loop featuring a high-pass transfer function to the output. The combination of the two injection points, the low- and the high-pass one, allows achieving wider modulation bandwidth. Obviously, the two-point injection technique to properly operate needs a good matching between the two injection paths. An alternative solution is based on the *pre-emphasis* of the digital modulation signal, this technique was originally exploited by Perrott *et al.* [5] to modulate at high rate the output frequency of a PLL. In practice, a pre-emphasis filter is introduced to approximate the inverse frequency response of the PLL. Like in the two-point injection scheme, good matching between pre-emphasis filter and inverse PLL response is required in this case.

The key issue of both approaches is therefore to guarantee matching between two different transfer functions within a certain degree. The least mean squares (LMS) algorithm has been already demonstrated to allow background calibration of building blocks variations and correction of mismatches [6], [7], and this makes digital phase-locked loops (DPLLs) the natural candidates to implement such systems. Unfortunately, in any PLL, the characteristics of the building blocks are not only unpredictable and dependent on process, voltage and temperature variations, but they are also nonlinear. Both the gain of a standard voltage controlled oscillator (VCO) and the gain of a digitally controlled oscillator (DCO) can substantially change over the tuning range. Even in structure that are optimized to this purpose it is difficult to maintain the gain variation below 5%, [2], while in standard structures the gain can change by 100% over the tuning range, [8]. A typical value of the gain error that can be considered is about 20% across the tuning range. This is usually not an issue in a PLL that works at a fixed output frequency, but it is relevant when the PLL is employed to frequency-modulate the carrier over a wide range. In that case, the calibration algorithm cannot track gain variations, giving rise to gain estimation errors.

In this paper, we extend the analytical comparison between the two-point injection and the pre-emphasis techniques which was introduced in [9] and [10]. In particular, we analyze the behavior and the performance of the two techniques in the presence of nonlinearity of the DPLL building blocks.

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The analysis will be based on a linear z -domain model of the modulators, where block nonlinearity will be modelled as gain estimation errors. The validity of this approach and the outcomes of theory will be numerically verified using an accurate time-domain behavioral model of a DPLL, which includes the nonlinearity of the building blocks. We will show that, while the two architectures have the same resilience to nonlinearity in the loop, the design of the phase detector is more relaxed in the two-point injection solution. To the best of the author's knowledge, a performance comparison of the pre-emphasis and the two-point injection methods and an analysis of the impact of nonlinearity in the two schemes have never been addressed in the open literature.

The paper is organized as follows: Section II discusses the requirements of the frequency modulator for FMCW radars and the basic trade-off between modulation bandwidth and phase noise in PLLs. It also reviews the practical limitations of state-of-the-art implementations and the two possible techniques to overcome them. In Section III, the theoretical analysis of both modulation enhancement techniques is carried out. Section IV describes the design of the DPLL including the two techniques and the simulation results verifying the proposed theory. Finally, the conclusions are drawn in Section V.

II. FREQUENCY MODULATOR FOR RADAR SYSTEMS

A. FMCW Radar Requirements

The accuracy of a radar system directly depends on the phase noise of the frequency modulator generating the ramp and on its linearity [11]. The phase noise of the modulator causes random frequency deviation in the transmitted chirp signal, which degrades the signal-to-noise ratio (SNR). In FMCW radars, the transmitted signal is also used to mix down the received signal to baseband. Thus, there is some degree of correlation between the phase noise of the transmitted and received signals that can be exploited to improve performance [12]. However, this correlation vanishes as the round-trip time of the transmitted signal increases. In practice, long range radars are typically sensitive to the phase noise above 1 MHz offset from the carrier [11]. This may set an upper limit to the bandwidth of the frequency synthesizer, used as frequency modulator. This limit depends both on the noise contributors within the PLL itself and also on the noise performance of the other building blocks of the radar, but typically the bandwidth does not exceed a value of about 500 kHz.

The linearity of the frequency ramp is affected by both linear and nonlinear distortion of the frequency modulator. The limited bandwidth of the modulator gives rise to linear distortion. To get a linearity error below 1% for a triangular chirp, the bandwidth of the modulator shall be larger than the 100th harmonic frequency of the modulation signal [13]. In the case of a saw-tooth chirp signal, the modulator bandwidth shall be much larger than that. In practice, to fulfill modulation speed and linearity requirements of modern radar systems, the modulator bandwidth should be pushed over several tenths of megahertz, which

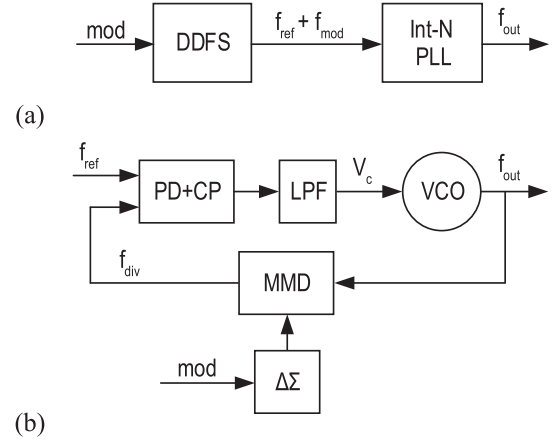


Fig. 1. Popular schemes of frequency synthesizers for direct FM modulation: (a) cascade of DDFS and PLL, (b) fractional- N PLL.

may be in contrast with phase noise optimization. Moreover, given also the limited bandwidths of commercially available modulators, concepts like gating time or gated FFT are applied today at the system level to reduce the impact of linear distortion [14].

B. PLL-Based Chirp Generator

Conventionally, chirp generators are based on high performance analog charge-pump PLLs with incorporated frequency modulation capability, often implemented in BiCMOS processes [15]–[20]. Figure 1 presents two architectures of PLL-based frequency synthesizer with frequency modulation incorporated in the reference or feedback path [15]–[18]. The architecture proposed in [15] consists of a direct digital frequency synthesizer (DDFS) which is followed by a PLL as depicted in Fig. 1(a). A DDFS provides a frequency modulated clock which is fed into the reference input of the PLL. In this configuration, the PLL does not require fractional- N resolution, which relaxes its design and improves noise and spur performance. However, low quantization noise and high linearity of the DDFS signal are obtained at the expense of large power consumption, in the order of few watts. Another typical PLL-based frequency modulator architecture employed in [16]–[18] is presented in Fig. 1(b). The modulation is implemented by means of a multi-modulus divider (MMD) driven by a digital modulator, in the feedback path of the PLL. In principle, the output frequency of the voltage-controlled oscillator (VCO) follows the linear sweep of the feedback divider ratio. Both the solutions in Fig. 1, however, feature the same key problem: In both architectures, the transfer function from the modulation signal *mod* to the output frequency is given by the input-to-output transfer function of a PLL. Thus, the modulation is subject to a low-pass frequency response and thus to a linear distortion, even assuming ideally linear building blocks. To limit the distortion introduced by the PLL, the maximum rate of the modulation signal should be restricted to a fraction of the closed-loop bandwidth of the PLL itself. One obvious solution to enable faster modulation rates is to widen the bandwidth of the PLL. Unfortunately, at constant reference frequency, this compromises loop stability, and, what

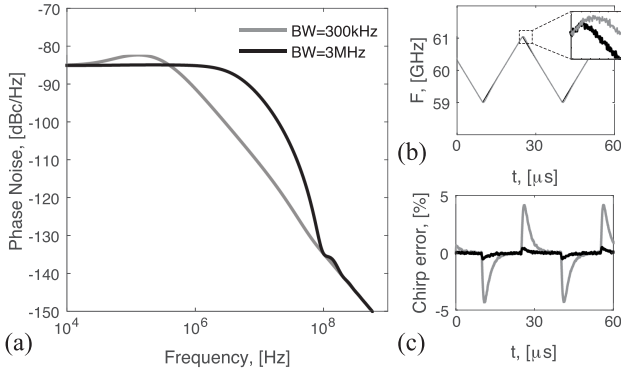


Fig. 2. Simulated results for a chirp generator implemented as a second-order fractional- N PLL with 300kHz and 3MHz loop bandwidth: (a) phase noise spectrum of the unmodulated PLL, (b) chirp signal, (c) chirp error.

is more important, may result in a non-optimal phase noise profile, which is critical for radar accuracy [11].

To quantify the trade-off existing between phase noise and linear distortion, we generated a chirp signal with a 60-GHz second-order fractional- N PLL with 100 MHz reference frequency. A triangular chirp with 30- μ s period (i.e. about 33 kHz frequency) and 2-GHz peak-to-peak amplitude was generated by modulating the division factor of the MMD. Two cases with the bandwidth of the PLL set to 300 kHz and 3 MHz were considered. Figure 2 shows, for the two bandwidth values, the phase noise spectrum, the output frequency modulation and the normalized chirp error, that is defined as:

$$\epsilon_{chirp} = \frac{\Delta f_{error}}{\Delta f_{pp}}, \quad (1)$$

where Δf_{error} is the deviation of the output frequency signal from the ideal chirp and Δf_{pp} is the peak-to-peak frequency range of the chirp.

The peak-to-peak chirp error is about 9% with the 300 kHz bandwidth, while it reduces to about 1% with the 3 MHz bandwidth. If the bandwidth is enlarged, however, the noise sources affecting the PLL forward path are less filtered and may become dominant both in term of spot noise and integrated jitter. Usually these contributors are the frequency divider, the phase detector and charge pump in analog PLL and the time-to-digital converter (TDC) in DPLL, but other noise sources may be important as well. Referring again to the example in Figure 2, the absolute jitter goes up from 250 to 650 fs, which exceeds the value of 590 fs reported in [2]. The spot phase noise at 1 MHz offset is about -85 dBc/Hz while a value of -90 dBc/Hz is considered appropriate, to relax the design of the other radar's building blocks. This analysis highlights the trade-off existing between phase noise and achievable chirp linearity in a conventional PLL architecture.

C. Wideband Modulation Schemes

The trade-off between bandwidth and noise can be overcome by exploiting one of the two techniques discussed in the following. In both cases, PLL bandwidth will be designed for optimum integrated phase noise. The first solution was proposed in the field of wireless communication in [21] and [22].

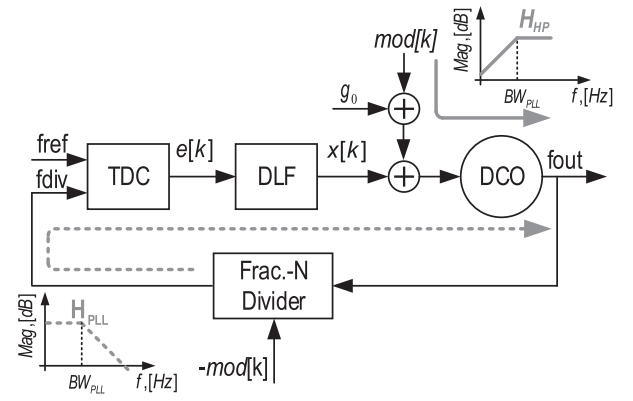


Fig. 3. Block diagram of a DPLL with the two-point injection technique [10].

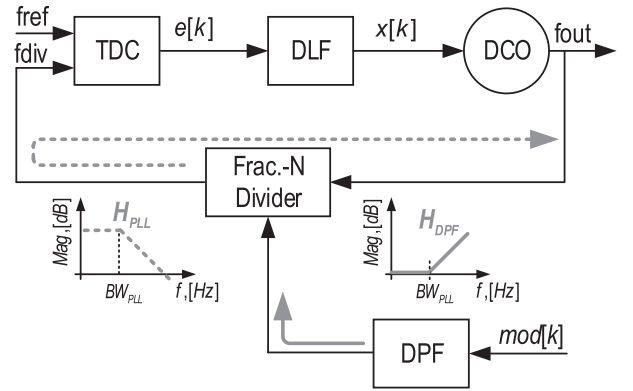


Fig. 4. Block diagram of a DPLL with the pre-emphasis technique [10].

The basic idea of the *two-point injection technique* implemented in a DPLL architecture is depicted in Fig. 3. This architecture employs a digital fractional- N frequency synthesizer with the modulation signal $mod[k]$ simultaneously injected via the feedback frequency divider and added to the output of the digital loop filter (DLF), $x[k]$. A TDC performs the phase detector function. Its output is the quantized phase difference between the reference and the divided clocks. The modulation technique utilizes the high-pass transfer function from DCO input to the PLL output as a second point of injection for the modulation signal. To match the level of the high-pass gain to the low-pass one, the modulation signal $mod[k]$ is multiplied by a constant g_0 at the high-pass injection point. The adoption of this technique for chirp generation was proposed in [2] and implemented in a 60-GHz TDC-based DPLL in a 65-nm CMOS process. More recently, in [3] and [4], the two-point injection technique was employed in a hybrid and a DTC-based PLL, respectively. Based on this technique, those works demonstrate chirp slopes up to 32 MHz/ μ s with linearity error lower than 1%.

An alternative solution already known from wireless communications is the *pre-emphasis* of the modulation signal. The essence of this modulation enhancement technique is depicted in Fig. 4. A digital pre-emphasis filter (DPF) shapes the modulation signal $mod[k]$ before entering the phase-locked loop to compensate for the PLL transfer function roll-off

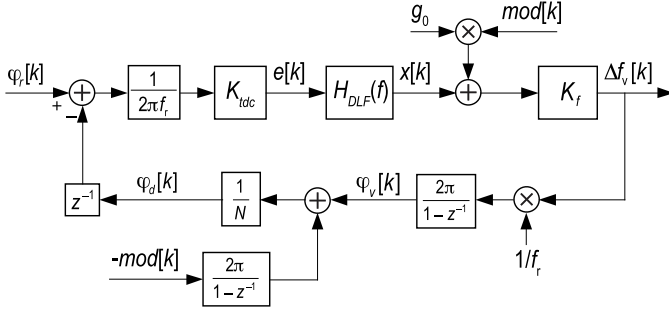


Fig. 5. Block diagram of the DPLL with the two-point injection technique.

beyond the bandwidth. More recent works on transmitters for wireless communications in [23]–[25], propose several calibration techniques for the accurate pole and zero placement in the DPF, further improving efficiency and robustness. However, to the best of the authors' knowledge, this technique has never been employed in the generation of chirp signals for radar systems and never compared at fundamental level with the alternative modulation technique, the two-point injection.

III. ANALYSIS OF THE MODULATION TECHNIQUES

A. Two-Point Injection

The simplified linear phase-domain model of a PLL that includes the two-point injection technique is presented in Fig. 5, where $\varphi_r[k]$, $\varphi_d[k]$ and $\varphi_v[k]$ represent the variable phase of the reference, divided and output clocks, respectively. The signal $\Delta f_v[k]$ is the variation of the DCO output frequency and f_r is the reference frequency. The transfer function of the DLF is the standard proportional-integral one $H_{DLF}(z) = \beta + \alpha/(1 - z^{-1})$.

The expression of the loop gain is given by

$$G_{loop}(z) = K \cdot H_{DLF}(z) \cdot \frac{z^{-1}}{1 - z^{-1}}, \quad (2)$$

where $K = K_f K_{tdc}/(N f_r^2)$, being N the feedback divider ratio, and K_{tdc} the TDC gain expressed in [s/LSB].

The modulation signal $mod[k]$ is simultaneously applied with inverted sign to the feedback path of the PLL as well as added to the output of the loop filter $x[k]$. Denoting as $mod(z)$ and $\Delta f_v(z)$ the z -transforms of the time-domain signal $mod[k]$ and the output frequency variation $\Delta f_v[k]$, and being G_{loop} the PLL loop gain, the transfer function from $mod(z)$ injected into the feedback to $\Delta f_v(z)$ is:

$$H_{lp}(z) = f_r \cdot \frac{G_{loop}}{1 + G_{loop}}, \quad (3)$$

that is the standard low-pass transfer function of a PLL multiplied by f_r . Denoting as K_f the DCO gain expressed in [Hz/LSB], the transfer function from the other point of injection, at the DLF output, is instead the following one:

$$H_{hp}(z) = \frac{g_0 K_f}{1 + G_{loop}}, \quad (4)$$

that is a high pass one.

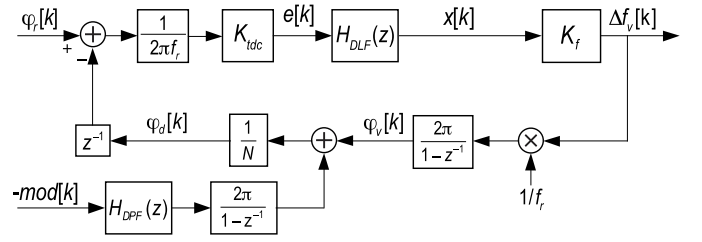


Fig. 6. Block diagram of the DPLL with the pre-emphasis technique.

If the value of the gain g_0 is exactly equal to f_r/K_f , the linear superposition of the two transfer functions results in an all-pass transfer function:

$$\begin{aligned} H_{mod,2P}(z) &= H_{lp}(z) + H_{hp}(z) \\ &= f_r \cdot \left(\frac{G_{loop}}{1 + G_{loop}} + \frac{1}{1 + G_{loop}} \right) = f_r. \end{aligned} \quad (5)$$

In practice, the DCO gain K_f in (4) is not only variable over process, temperature and voltage (PVT), but also dependent on the output frequency, given the nonlinearity of a typical DCO tuning characteristic. Thus, this technique requires an accurate estimation of g_0 (to guarantee that it is always equal to f_r/K_f), which is typically done by a dedicated calibration circuitry. As discussed in detail in [7] the error between g_0 and f_r/K_f gives rise to a digital signal at the TDC output that is correlated to the modulation signal. Exploiting this correlation with a least-mean square (LMS) algorithm it is possible to estimate the value of the gain g_0 . Any discrepancy between the ideal value of g_0 and its estimate would appear as an additional phase error at the TDC input and distort the output chirp signal.

B. Digital Pre-Emphasis

The pre-emphasis consists in introducing a digital filter (digital pre-emphasis filter, DPF) into the modulation signal path, which compensates for the roll-off of the PLL transfer function. If the transfer function of the DPF were the exact inverse of the PLL transfer function, the total transfer function of their cascade would be an exact all-pass one. The linear model of this architecture is depicted in Fig. 6, the overall transfer function from the modulation signal to the output frequency deviation is:

$$H_{mod,DPF}(z) = H_{DPF}(z) \cdot H_{lp}(z), \quad (6)$$

where $H_{DPF}(z)$ is the transfer function of the DPF, while $H_{lp}(z)$ is the PLL transfer function given in (3). The DPF transfer function can be derived by inverting the PLL transfer function, so that the resulting $H_{mod,DPF}(z) = f_r$. Hence, we obtain

$$\begin{aligned} H_{DPF}(z) &= \frac{1 + G_{loop}(z)}{G_{loop}(z)} \\ &\approx \frac{1 + [K(\alpha + \beta) - 2]z^{-1} + (1 - K\beta)z^{-2}}{K(\alpha + \beta) - K\beta z^{-1}}. \end{aligned} \quad (7)$$

The above expression is an approximation of the inverse transfer function of the PLL, since a z^{-1} delay has been

removed from the denominator to guarantee the causality of the filter. The resulting filter is a second-order IIR one, whose coefficients depends on a dispersed parameter, K , that in turn is proportional to the product of the DCO gain, K_f , and TDC gain, K_{tdc} . Those parameters are not only subject to PVT variations, but also dependent on the variations of bias condition, because of the practical nonlinear characteristics of DCO and TDC. To avoid that any mismatch or drift could cause a degradation of the modulation accuracy, a method for automatic matching between the two transfer functions, the PLL and DPF one, should be implemented in this case as well. The calibration circuit, as presented in [23], can be either tuned to match the inverse transfer function of the DPF filter leaving the PLL transfer function unchanged or, vice versa. According to [23], in which an analog PLL is modulated, the latter technique is to be preferred, since a fixed DPF filter can be efficiently implemented (e.g. with a lookup table) while to adjust the PLL bandwidth is usually possible via tuning, for example, a charge pump current. Another work adopting this approach is [24]. On the other hand, in [25] the DPF transfer function is tuned via a single digital coefficient, the PLL being an analog one. In this case a replica of the RC loop filter and an analog-to-digital converter are required.

Of course the two choices are possible also in the DPLL. For instance a digital parameter can be used to regulate its bandwidth, leaving the DPF fixed. This parameter can scale the TDC output, see one example in [26], and the computational effort is equivalent to the estimation of g_0 in the two-point technique.

C. Impact of Gain Errors

In both the cases so far analyzed, any inaccuracy in DCO and TDC gain estimates, $\hat{K}_f$ and $\hat{K}_{tdc}$ would alter $H_{mod}(z)$ from its ideal value and give rise to the following frequency error:

$$\Delta f_{error} = mod(z) \cdot [f_r - H_{mod}(z)], \quad (8)$$

where $H_{mod}(z)$ is the transfer function from $mod(z)$ to $\Delta f_v(z)$, given by either (5) or (6) depending on the modulation technique considered.

Intuitive considerations may suggest that the error made in the estimation of K_f and K_{tdc} is suppressed by the loop to some extent. However, how this error impacts the output Δf_{error} is not obvious and has never been addressed in the literature. In the following, the analysis of chirp linearity errors in the presence of gain errors will be presented for both modulation techniques.

At first, the analysis is carried out for the two-point injection technique. Replacing the DCO gain in (4) with the estimate $\hat{K}_f$ and computing again the expression of $H_{mod}(z)$, we derive the following equation:

$$\begin{aligned} H_{mod,2P}(z) &= f_r \cdot \left(\frac{K_f}{\hat{K}_f} \cdot \frac{1}{1 + G_{loop}(z)} + \frac{G_{loop}}{1 + G_{loop}(z)} \right) \\ &= f_r \cdot \frac{1 + (1 + \epsilon_{K_f}) \cdot G_{loop}(z)}{(1 + \epsilon_{K_f}) \cdot [1 + G_{loop}(z)]}, \end{aligned} \quad (9)$$

where the last expression follows after introducing the relative gain error ϵ_{K_f} , such that $\hat{K}_f = K_f \cdot (1 + \epsilon_{K_f})$.

The normalized chirp error for the two-point injection scheme can be obtained from (1) and (8) by replacing the expression of H_{mod} achieved in (9):

$$\begin{aligned} \epsilon_{chirp,2P}(z) &= \frac{mod(z) \cdot [f_r - H_{mod,2P}(z)]}{\Delta f_{pp}} \\ &= \frac{mod(z) \cdot f_r}{\Delta f_{pp}} \cdot \frac{\epsilon_{K_f}}{1 + \epsilon_{K_f}} \cdot \frac{1}{1 + G_{loop}(z)}. \end{aligned} \quad (10)$$

Equation (10) reveals that, in the presence of DCO gain estimation error, the spectrum of the chirp error is given by the spectrum of the modulation signal after high-pass shaping with dominant pole located at around the closed-loop bandwidth of the PLL [the last factor in (10)]. This means that the PLL is able to reject chirp errors induced by the gain error, as long as the modulation speed is slow enough with respect to the speed of the closed loop. If, instead, a portion of the spectrum of $mod[k]$ falls outside PLL bandwidth, the loop plays no role and the error propagates to the output causing chirp distortion.¹

The situation is instead somewhat different in the pre-emphasis approach. In this case, the same analysis is performed by exploiting the linear model in Fig. 6. The $H_{mod}(z)$ transfer function is described by the following equation:

$$\begin{aligned} H_{mod,DPF}(z) &= \hat{H}_{DPF}(z) \cdot H_{lp}(z) \\ &= f_r \cdot \frac{1 + \hat{G}_{loop}(z)}{\hat{G}_{loop}} \cdot \frac{G_{loop}}{1 + G_{loop}(z)}, \end{aligned} \quad (11)$$

where we introduced the estimated loop gain, $\hat{G}_{loop}$. The latter is proportional to the product $\hat{K} = K \cdot (1 + \epsilon_K)$, where ϵ_K is the relative error in the estimation of K . Hence, the estimated loop gain can be written as follows:

$$\begin{aligned} \hat{G}_{loop} &= \hat{K} \cdot H_{DLF}(z) \cdot \frac{z^{-1}}{1 + z^{-1}} \\ &= G_{loop}(z) \cdot (1 + \epsilon_K). \end{aligned} \quad (12)$$

As K is proportional to the product of DCO and TDC gains, $K_f \cdot K_{tdc}$, it follows that the total error is approximately given by the sum of the relative errors of the two gains: $\epsilon_K \approx \epsilon_{K_f} + \epsilon_{K_{tdc}}$, assuming $\epsilon_{K_f} \epsilon_{K_{tdc}} \ll \epsilon_{K_f} + \epsilon_{K_{tdc}}$.

The normalized chirp error for the pre-emphasis scheme, in the presence of gain errors, is obtained from (1) and (8), using the transfer function in (11):

$$\begin{aligned} \epsilon_{chirp,DPF}(z) &= \frac{mod(z) \cdot [f_r - H_{mod,DPF}(z)]}{\Delta f_{pp}} \\ &= \frac{mod(z) \cdot f_r}{\Delta f_{pp}} \cdot \frac{\epsilon_K}{1 + \epsilon_K} \cdot \frac{1}{1 + G_{loop}(z)}. \end{aligned} \quad (13)$$

Comparing (10) and (13), we conclude that the error in the estimation of the DCO gain is suppressed in the very same way in both modulation techniques.

¹More rigorously, the loop gain, G_{loop} , in (10) should be slightly different from the nominal one, since it is also affected by both DCO and TDC gain errors, ϵ_{K_f} and $\epsilon_{K_{tdc}}$. This correction, however, implies only a negligible variation in the loop gain value, and does not alter appreciably the final result given in (10).

From the presented analysis, we can also get a valuable insight. In the two-point injection scheme, the loop is resilient to gain mismatches between the two injection paths, as long as the modulation signal is within the loop bandwidth. This is because a feedback loop, with loop gain much larger than one, tends to suppress any gain variations in the forward path. In the pre-emphasis scheme, the resilience to gain errors is given by the fact that for loop gain much larger than one, the transfer function of the pre-emphasis filter does not significantly change even in the presence of gain errors. By contrast, fast variations of the modulation signal (beyond the loop bandwidth) leak to the output in the presence of gain errors and give rise to modulation errors in both schemes. Thus, to achieve fast chirps with large modulation bandwidths, an automatic calibration is required in both cases.

Note that the modulation error for the two-point injection scheme is only sensitive to DCO gain variations, while for the pre-emphasis technique, modulation accuracy is intrinsically affected by both DCO and TDC gain errors. The key point to highlight is that the analysis presented here is also suitable to predict the impact of blocks nonlinearity. Consider as an example the DCO characteristic, typically the most nonlinear one in a PLL. An automatic estimation of its gain, K_f , will provide an average value of the slope of the tuning characteristic. During the frequency sweep, the slope of the characteristic changes. This variation can be equivalently seen as the addition of a variable error ϵ_{K_f} to the estimate. Since slow gain errors are suppressed by the loop, we may expect that only the fast components of this error, arising at the turn-around points of the frequency signal, will alter the frequency-modulated output. In other words, we can estimate the chirp error induced by circuit nonlinearity by evaluating ϵ_{K_f} at the turn-around points and using either (10) or (13).

D. TDC Dynamic Range

A significant difference between the two bandwidth enhancement techniques is the magnitude of phase error they produce in the presence of gain errors. This is a relevant aspect because the TDC dynamic range needs to accommodate this additional term, avoiding saturation. A larger dynamic range typically demands for larger power consumption.

Based on the simplified model in Fig. 5, the phase error at TDC output for the two-point injection scheme is as follows:

$$e_{2P}(z) = \left(\frac{g_0 K_f}{f_r} - 1 \right) \cdot \frac{G_{loop}(z)}{1 + G_{loop}(z)} \cdot \frac{mod(z)}{K_f H_{DLF}(z)}. \quad (14)$$

Hence, in the case of ideal matching, i.e. $g_0 = f_r/K_f$, the TDC output, as well as its input, is always zero. This should not produce the wrong conclusion that the loop is open; it only means that the modulated signal does not appear at the TDC input. As a matter of fact, in the previous subsection we have noticed that the system is able to reduce the impact of the error within the loop bandwidth.

We can compute the phase error induced by the modulation signal in the pre-emphasis case as well. In that case, we obtain:

$$e_{DPF}(z) = H_{DPF}(z) \cdot \frac{G_{loop}(z)}{1 + G_{loop}(z)} \cdot \frac{mod(z)}{K_f H_{DLF}(z)}. \quad (15)$$

Assuming that the pre-emphasis filter perfectly cancels the second factor in (15), the remaining term is as follows:

$$e_{DPF}(z) = \frac{mod(z)}{K_f H_{DLF}(z)}. \quad (16)$$

The same residual error would be also obtained in a standard PLL configuration (without pre-emphasis, $H_{DPF} = 1$, assuming a large loop gain, $G_{loop} \gg 1$).

The above expression highlights a clear advantage of the two-point injection technique over the pre-emphasis technique. In fact, (16) highlights that the low frequency components of the signal is high-pass shaped. Since the loop filter transfer function H_{DLF} can be approximated, at low frequency, to that of a pure integrator, the error signal $e[k]$ in the same frequency range is proportional to the first difference of $mod[k]$. Consider, for instance, a triangular modulation of the output frequency. In that case, the first difference of $mod[k]$ is a square-wave sequence, that periodically changes its sign. Using the pre-emphasis technique, the TDC input (and output as well) has to follow it, going swinging between a negative and positive value, in the case of perfect matching of the coefficients.

Instead, in the case of two-point injection scheme, the TDC signal is ideally zero. Even in the practical case, where gains are mismatched, the dynamic range required to the TDC for the two-point case is much more relaxed than for the pre-emphasis case. Simulations in the following section will show this difference. It is worth mentioning that for a signal with even larger derivative step, such as a saw-tooth chirp signal, the advantage of the two-point injection scheme is more significant.

IV. SIMULATION RESULTS

A. Architecture of the Digital PLL

To assess the validity of the proposed analysis, two digital PLLs with direct frequency modulation (FM) have been designed. The first one adopts the two-point scheme, described in Fig. 3, and the second one adopts the pre-emphasis technique in Fig. 4.

The architecture of the DPLL is similar to the one proposed in [6]. The reference frequency is 100 MHz. Phase detection is implemented by means of a TDC with mid-tread quantization characteristic and resolution of 4 ps. The DCO has a center frequency of about 60 GHz and a 12-bit digital word controlling its frequency with resolution K_f of 1 MHz/LSB. Both TDC and DCO resolutions are practical values achievable in a modern CMOS process. To further improve frequency resolution, the DCO is driven by a first-order digital $\Delta\Sigma$ modulator. The digital loop filter (DLF) is implemented as a proportional/integral filter with programmable proportional gain β and integrator coefficient α . The fractional- N divider is a cascade of a divide-by-4 block and a MMD driven by a $\Delta\Sigma$ modulator. The quantization noise of the above $\Delta\Sigma$ modulator is cancelled at the output of the TDC, using the LMS technique proposed in [6]. The DPLL parameters are summarized in Table I.

The time domain models of the two DPLLs are presented in Fig. 7. Both models are developed across two clock domains,

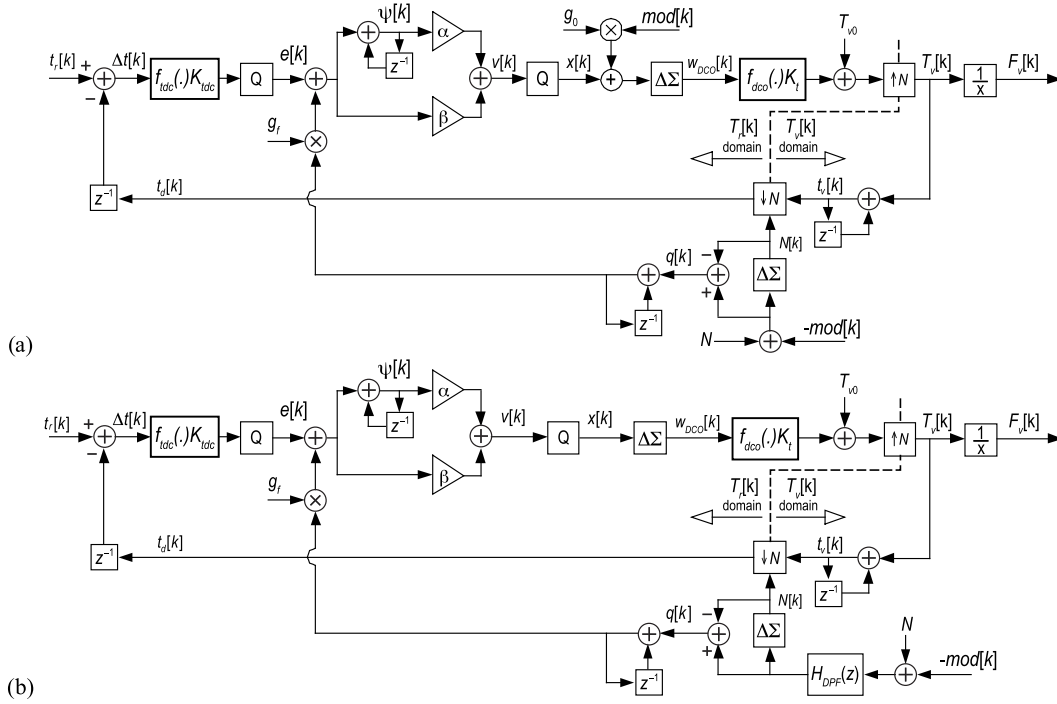


Fig. 7. Time-domain behavioral models of a DPLL with (a) two-point injection and (b) digital pre-emphasis techniques.

TABLE I
PLL DESIGN PARAMETERS

Parameter	Symbol	Value	Unit
DCO gain	K_f	1	MHz/LSB
TDC gain	K_{tdc}	250	LSB/ns
DCO nom. frequency	f_{v0}	60	GHz
Reference frequency	f_r	100	MHz
DLF proportional gain	β	6	-
DLF integral gain	α	0.012	-
Reference phase noise	$\mathcal{L}_{ref}$	-145	dBc/Hz
DCO phase noise at 1MHz offset	$\mathcal{L}_{dco}$	-95	dBc/Hz

namely the reference and the DCO ones, whose periods are $T_r[k]$ and $T_v[k]$, respectively [27]. The clock domain transitions are performed by interpolation and decimation functions. The quantization introduced by TDC and DCO blocks are modelled by quantizer functions (the Q blocks). Nonlinearity of TDC and DCO conversion characteristics is modelled by a nonlinear functions $f_{tdc}(\cdot)$ and $f_{dco}(\cdot)$, respectively. The model also accounts for white phase noise of the reference signal and for $1/f^2$ phase noise of the DCO signal.

The simulated phase noise spectrum $\mathcal{L}(f)$ of the DPLL, for a constant value of the frequency control word $FCW = 600.23$, is plotted in Fig. 8. In the same graph, the theoretical contribution to phase noise induced by reference and DCO phase noise, by TDC quantization noise and DCO truncation noise (DCO_Q). The latter is significantly below the other contributions. The bandwidth of the PLL is set to around 400 kHz, which minimizes the absolute jitter to an RMS value of 248 fs.

B. Linear Distortion

To assess the effectiveness of the two-point injection scheme and the pre-emphasis technique and quantify their

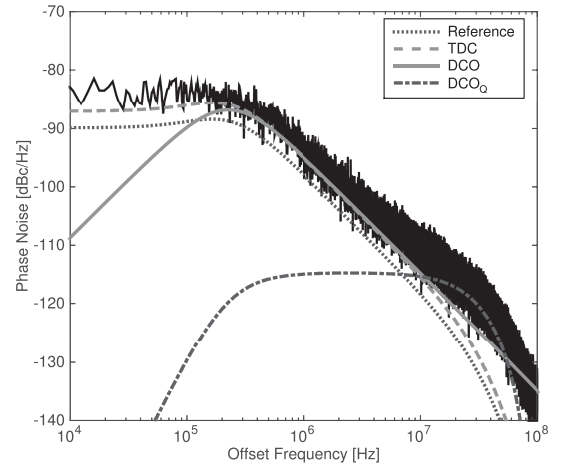


Fig. 8. Simulated phase noise of the unmodulated DPLL and theoretical contribution of the four contributors: reference and DCO phase noise, TDC quantization and DCO truncation noise (DCO_Q).

performance, the models in Fig. 7 along with a similar DPLL without bandwidth enhancement techniques were simulated. In the three cases, a triangular chirp with peak-to-peak modulation bandwidth Δf_{pp} of 3 GHz and 30- μ s period, and an ideally linear DCO characteristic has been used. For the bandwidth enhancement techniques, the DCO and TDC gain estimations, $\hat{K}_f$ and $\hat{K}_{tdc}$ are assumed to be ideal with no error and no quantization.

The results for these three ideal cases are plotted in Figs. 9(a)–9(c). Figure 9(a) shows the value of the DPLL output frequency superimposed to the ideal one, the chirp error defined in (1), and the error at TDC output, for the plain DPLL case. The DPLL achieves peak chirp error of about $\pm 4\%$, evaluated along 100% of the modulation period.

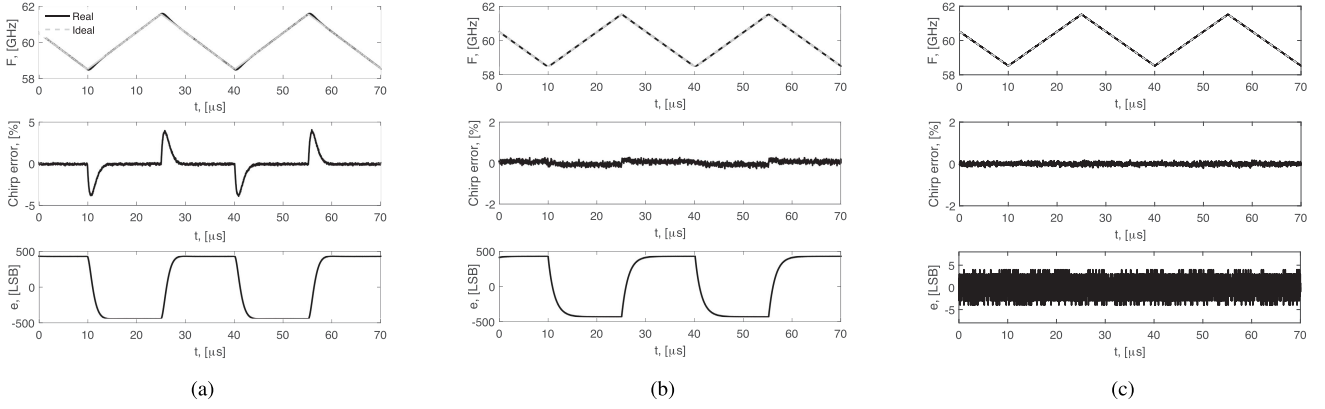


Fig. 9. Simulated carrier frequency, chirp error and TDC output for triangular chirp with (a) no bandwidth enhancement technique, (b) the pre-emphasis technique (no gain error), (c) the two-point injection technique (no gain error).

Such a performance is insufficient for most of radar applications. This behavior is caused by the limited bandwidth of the DPLL, which induces a transient after turn-around points of the triangular waveform. As evident from the transient of the chirp error, error below 1% is achieved only after about $5 \mu\text{s}$ from turn-around points. Thus, around 30% of the ramp-up and ramp-down of the triangular wave shall be discarded by the radar systems, which results in a significant degradation of radar performance. The last subplot shows the output of the TDC which approximately follows the derivative of the modulation signal, as discussed in Sect. III-C. The range of this error is about ± 500 LSB.

Figure 9(b) shows the simulated results obtained when the pre-emphasis filter, $H_{DPF}(z)$, is employed in the feedback path. Chirp linearity has greatly improved, leading to a peak error of $\pm 0.15\%$. The range of the error detected by the TDC is unchanged (about ± 500 LSB peak) with respect to the plain DPLL, which is in agreement with the discussion in Sect. III-D.

Finally, we simulated the DPLL with the two-point injection of the modulation signal. Figure 9(c) shows the achieved chirp error which is lower than $\pm 0.15\%$, as we obtained for the pre-emphasis. As expected from the above-presented analysis, the results obtained from the two-point injection technique are well in line with the performance of the pre-emphasis technique. Instead, as anticipated in Sect. III-D, the modulation signal is suppressed at TDC input and the TDC spans a significantly lower range (about ± 4 LSB peak against ± 500 LSB peak). The modulation signal is practically suppressed at TDC input and the residual error consists of random noise and $\Delta \Sigma$ quantization noise.

Considering the triangular FM signal with $30\text{-}\mu\text{s}$ period and $200\text{-MHz}/\mu\text{s}$ slope, the required number of TDC bits is, therefore, 10 bits for the plain DPLL and for the pre-emphasis modulation scheme, while it can be reduced to only 3 bits for the two-point injection scheme.

C. Nonlinear Distortion

A second set of simulations aims to quantify the impact of the error in the estimation of DCO and TDC gains in the two bandwidth enhancement techniques, and compare the

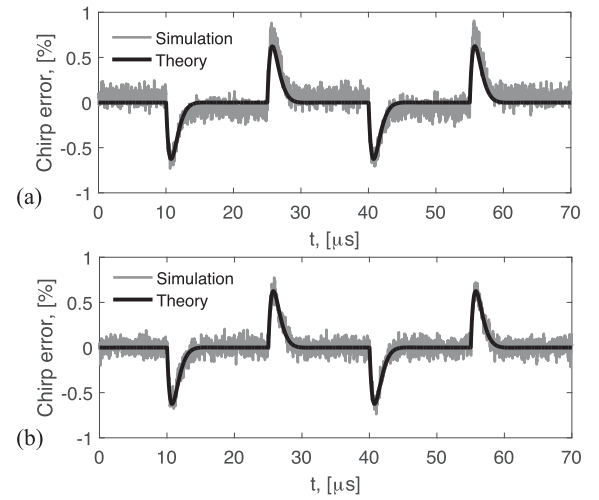


Fig. 10. Theoretical and simulated chirp error with 20% DCO estimation gain error for (a) the pre-emphasis and (b) the two-point injection techniques.

outcomes of the time-domain models in Fig. 7 to the theoretical results achieved in (10) or (13). In both systems, we accounted for an error, $\epsilon_{K_f} = 0.2$, in the estimate of DCO gain. The chirp error for the pre-emphasis and the two-point injection schemes are plotted in Fig. 10, along with the theoretical result obtained using the transfer function in (10). The simulated results are in very well agreement with theory and show that the simple linear models in Fig. 5 and 6 provide a good estimation of the chirp error. Moreover, this simulation results confirm that both enhancement techniques, in the presence of DCO gain error, achieve approximately the same chirp error (i.e., about 1.4% peak-to-peak error), as it was predicted in the proposed analysis. Similar outcome is obtained, when the same gain estimate error is imposed to the estimation of K_{Tdc} in the pre-emphasis case.

In a third set of simulations, we evaluated the behavior of the two bandwidth enhancement techniques in the presence of nonlinearity in the loop. In the models of the two DPLLs, we accounted for DCO nonlinearity by introducing the nonlinear $f_{dco}(\cdot)$ function.

In general, the shape of the tuning characteristic depends on the particular implementation either of the varactor

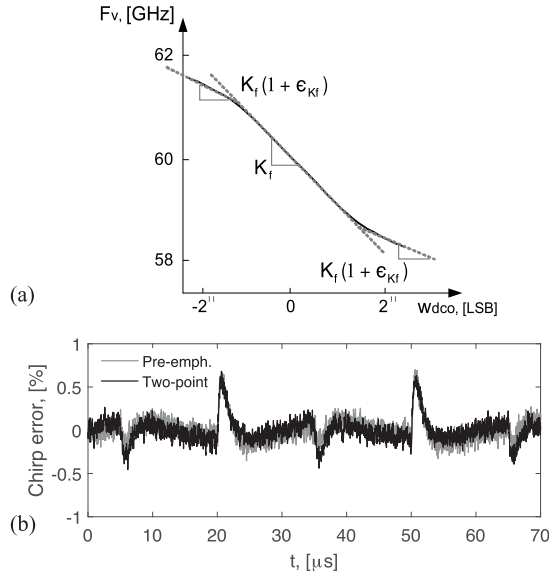


Fig. 11. Simulated chirp with $f_{dco1}(\cdot)$ DCO characteristic: (a) Sketch of DCO nonlinear characteristic, (b) chirp error for the pre-emphasis and the two-point injection techniques.

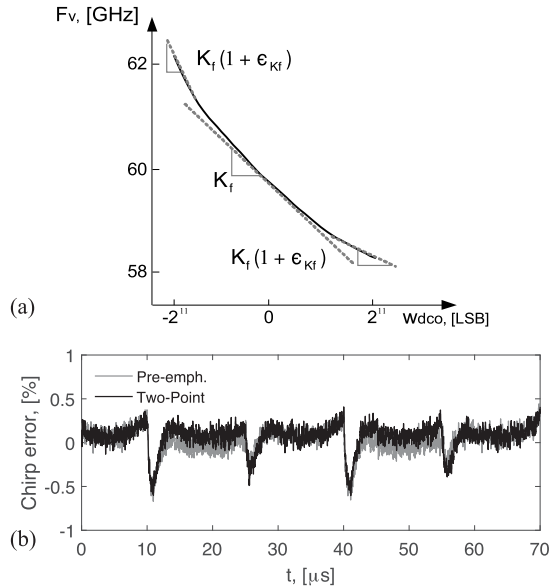


Fig. 12. Simulated chirp with $f_{dco2}(\cdot)$ DCO characteristic: (a) Sketch of DCO nonlinear characteristic, (b) chirp error for the pre-emphasis and the two-point injection techniques.

(for a VCO) or of the capacitive bank (for a DCO). Very often a mixed solution is adopted in which a coarse frequency tuning is performed by switching some capacitors, while the fine tuning is performed by a varactor. As case studies, we therefore considered two different nonlinear tuning characteristics. The first one, $f_{dco1}(\cdot)$, is representative of a gain sensitivity decreasing monotonically. The second one, $f_{dco2}(\cdot)$, is a saturating tuning curve and it is representative of a gain sensitivity having a maximum at the center of the tuning range.

In both characteristics, the integral nonlinearity (INL) is considered to be about 100 LSB at the boundaries of the used range. As a result, in the first case, the first derivative

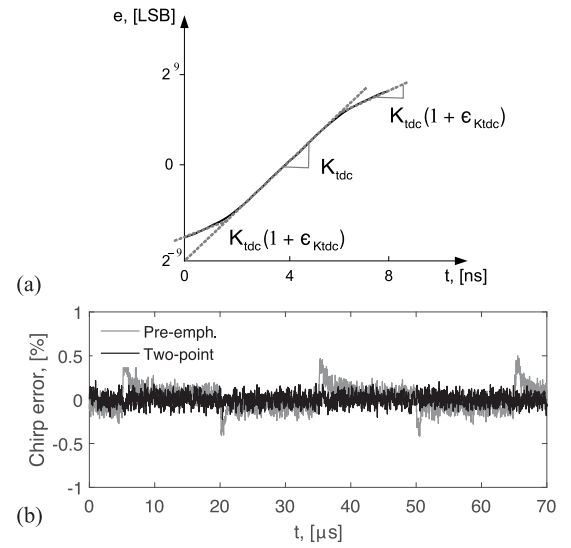


Fig. 13. Simulated chirp with nonlinear TDC characteristic: (a) Sketch of TDC nonlinear characteristic, (b) chirp error for the pre-emphasis and two-point injection techniques.

is about 20% lower at the boundaries of the range with the respect to the center. Thus, if DCO gain is estimated at center of the characteristic, a gain estimation error $\epsilon_{K_f} = 0.2$ will affect the boundaries of the characteristic, or, in other words, the peaks of the triangular frequency modulation. This case is schematically shown in Fig. 11(a). The chirp error obtained from numerical simulations is shown in Fig. 11(b). The peak-to-peak chirp error is about 1.2% for both bandwidth enhancement schemes, which is close to the error magnitude obtained in the linear case with 20% DCO gain error. This demonstrates that the impact of nonlinearity can be at first order accounted for by the simple linear models in Fig. 5 and 6, assuming a gain estimation error, and it also demonstrates that the two bandwidth enhancement techniques have the same resilience to loop nonlinearity.

In the second case, whose nonlinear characteristic is sketched in Fig. 12(a), the first derivative is about 20% higher in magnitude at the high-end of the characteristic and about 20% lower in magnitude at low end, with respect to the average slope. So, we have a relative error ϵ_{K_f} at the high end, and $-\epsilon_{K_f}$ at the low-end. It is interesting to note that while the peaks of the chirp error in Fig. 11(b) alternatively change their sign, the peaks of the chirp error in Fig. 12(b) are always negative. This behavior can be predicted from equations (10) and (13). In fact, in general, the gain error weights a sequence of peaks having alternate signs, since they correspond to the peaks of the triangular waveform. If this pulse train is weighted by an error sequence with same sign, such as the one originating from the cubic nonlinearity, the peaks of the chirp error will alternate their polarity. For the nonlinearity in Fig. 12, instead, the gain errors at the two extremes have opposite signs. So, the sequence of chirp errors is somewhat “rectified”.

Note that, according to this picture, in this second case it is impossible to obtain the sequence of the rectified peaks from (10) and (13) since with a linear model it is not possible

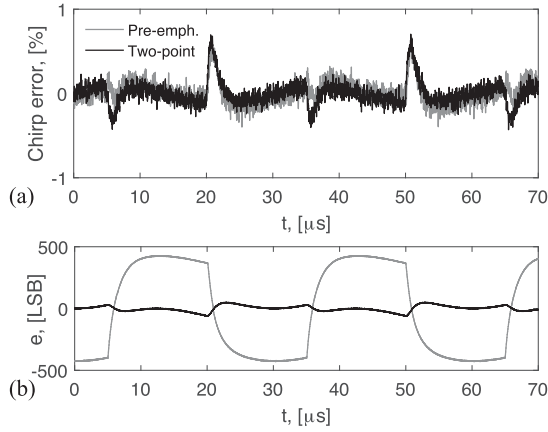


Fig. 14. Simulated chirp error with DCO characteristic from Fig. 11(a) and TDC characteristic from Fig. 13(a) and the resulting TDC output.

to have a different (linear) gain in different points of the characteristic. However the peak magnitude and shapes are well matched by the prediction of the linear model. This demonstrates again the effectiveness of the presented approach to evaluate the impact of loop nonlinearity. This is also a further confirmation that the two architectures behave in the same way in the presence of nonlinearity. While from the practical applications standpoint, the chirp sequence in Fig. 12(b) is neither beneficial nor detrimental with respect to the case in Fig. 11(b), the two cases demonstrate that equations (10) and (13) work properly.

The last sets of simulations aim to verify the impact of the TDC nonlinearities. The TDC odd characteristic in Fig. 13(a), which corresponds to $f_{ide}(\cdot)$, has been therefore introduced in the behavioral model, with a relative gain error of 20% at the extremes. The simulated chirp error in Fig. 13(b) shows that only the pre-emphasis technique is sensitive to this error. Finally, in Fig. 14 both the DCO and TDC nonlinearities are introduced. In addition to the chirp error this figure also shows the TDC error for the two different modulation techniques. The range required to the TDC for the two-point technique is, in this case, about one order of magnitude smaller with respect to the pre-emphasis case. This can be derived, at the first order, still comparing (14) and (16). Assuming a loop gain much larger than one, it is easy to derive that for a relative error of 20%, the TDC error obtained from (14) for a two-point modulator is no longer zero, but it is one order of magnitude smaller than the one from (16).

V. CONCLUSIONS

In this paper, we investigated architectures for the generation of a highly-linear chirp signal to be used in high-resolution FMCW radar systems, comparing the pre-emphasis and the two-point modulations technique applied to a DPLL. An analysis of the sensitivity to gain calibration errors of both architectures has been also carried out. This analysis can be also employed to derive the chirp error in the two architectures in the presence of nonlinearity of the building blocks. From this standpoint, the two modulation techniques are equivalent. However, the two-point injection architecture is sensitive only

to variations of DCO gain, while the pre-emphasis is also affected by TDC errors. The two-point injection technique also significantly relaxes the dynamic range requirement of the TDC block with respect to the other approach, which implies either better linearity or reduced power consumption of the TDC. The analytical results have been verified with an accurate time-domain model simulations of the proposed DPLL with different nonlinear characteristics.

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